PREDICTING CREDIT RISK USING SUPERVISED CLASSIFICATION

A COST-SENSITIVE EVALUATION ON THE GERMAN CREDIT DATASET

Research Question:

Can we predict credit risk and does the asymmetric cost of misclassification change which model a bank should prefer?

BY : MOHAMMED ADNAN

COMP4030 - Data Science with Machine Learning | University of Nottingham | 2026

Introduction

Asymmetric Cost

Approving a bad loan costs 5× more than rejecting a good applicant. The UCI dataset formalises this through an official cost matrix.

Accuracy is Misleading

70/30 class imbalance - predicting everyone as Good gives 70% accuracy without learning anything useful from data.

Goal

Compare 4 classifiers under cost-aware evaluation and find which model minimises total financial cost for a bank.

UCI Cost Matrix

	Pred. Good	Pred. Bad
Act. Good	✓	Cost × 1
Act. Bad	Cost × 5	✓

Misclassifying a Bad applicant as Good is 5× more costly - making recall and total cost more important than accuracy for this problem.

Dataset & EDA

1,000
Applicants
loan records

20→17
Features
after selection

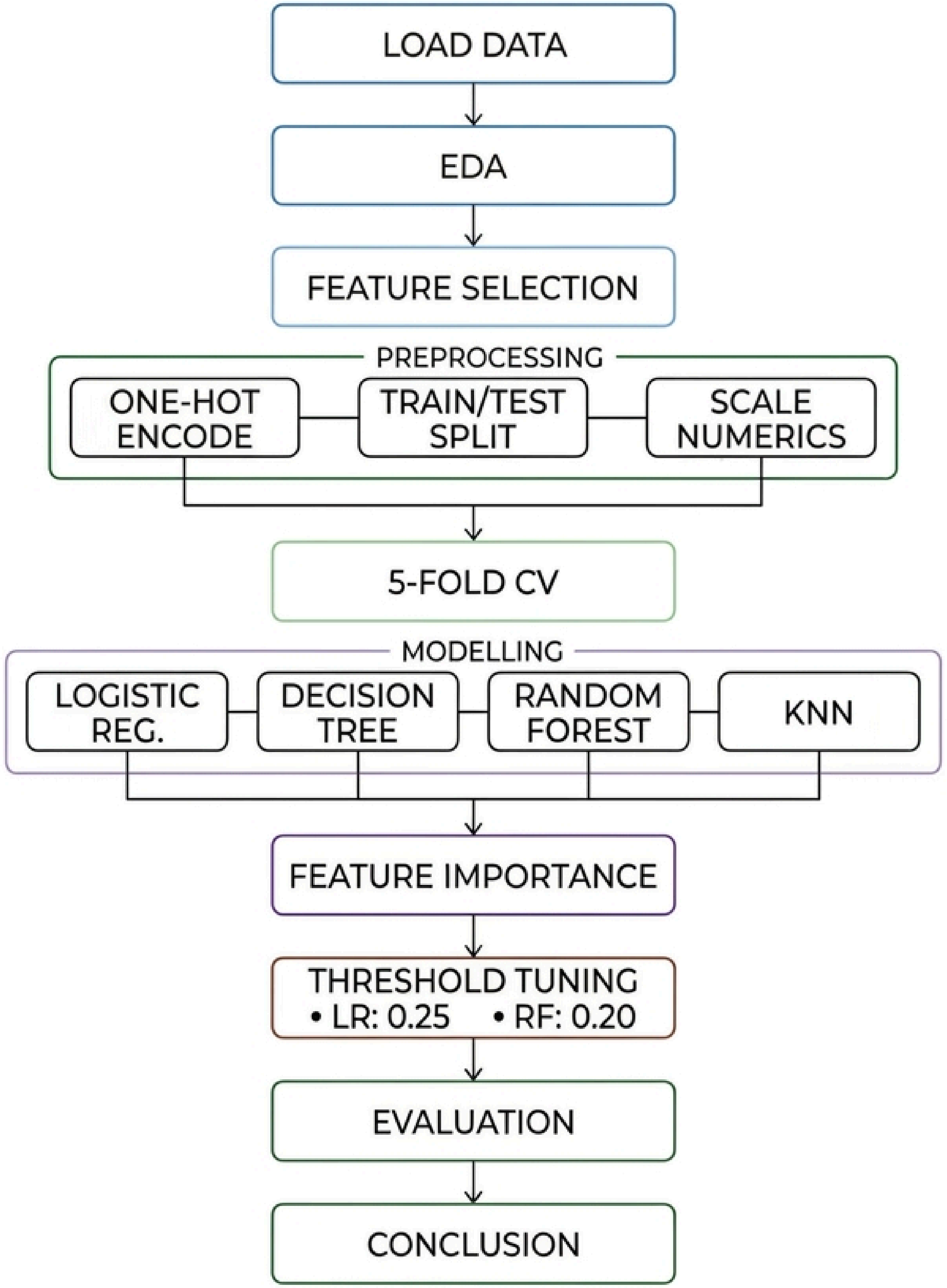
70/30
Class Split
Good vs Bad

Zero
Missing Values
clean dataset

Key Observations

- 7 numerical + 13 categorical features
- 3 features removed : own_telephone (var 0.0006), num_dependents (var 0.13), residence_since (var 1.22)
- checking_status and credit_history : highest bad rate variance, most discriminative
- 70/30 imbalance : F1 score used as primary metric
- Official UCI cost matrix: FN × 5, FP × 1

Methodology



Model Results - Test Set Performance & CV Stability

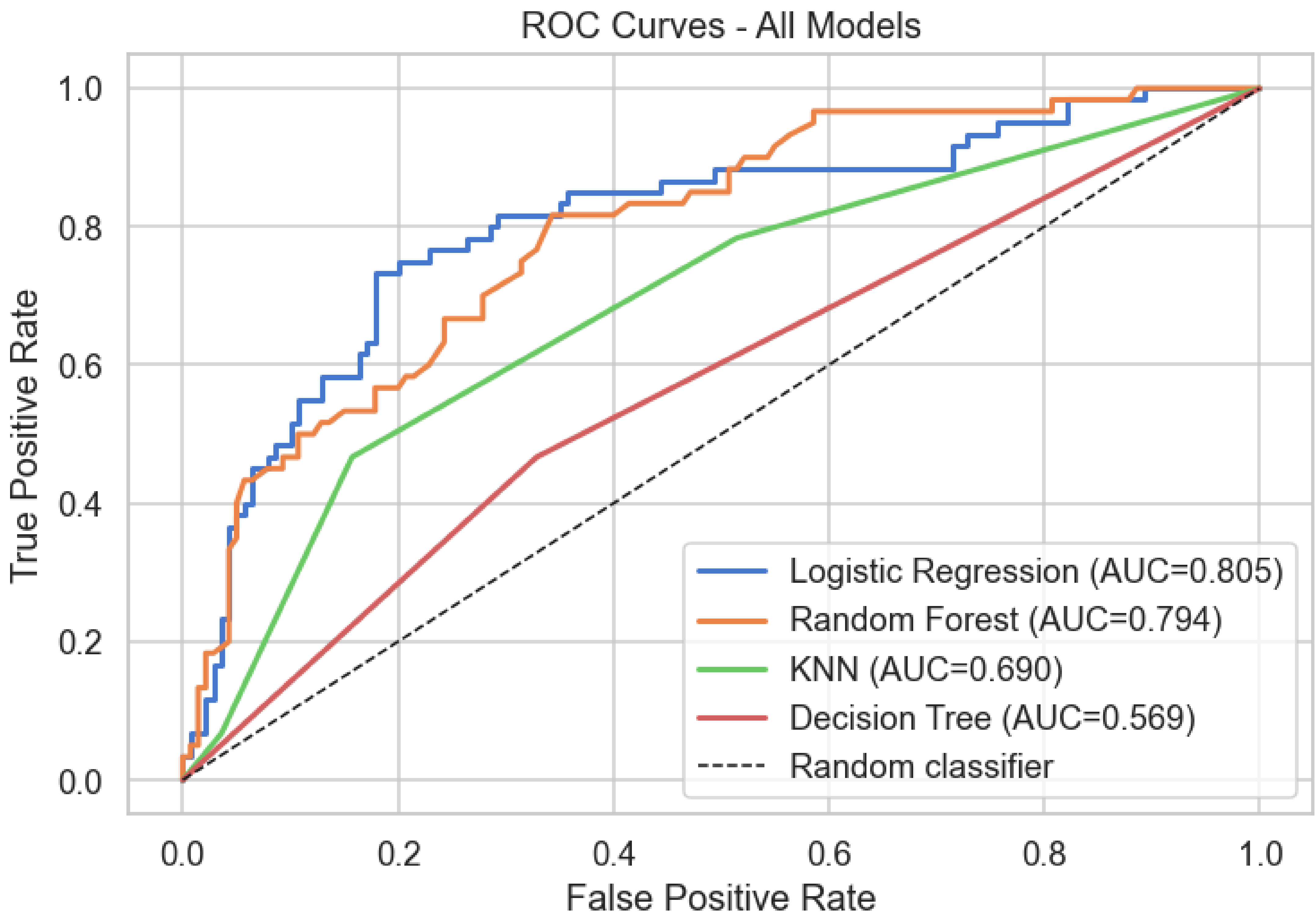
Model	Accuracy	F1 (Bad)	Recall	Precision	ROC-AUC	Total Cost
Dummy Classifier	0.7	**0**	**0**	0	0.5	**300**
Logistic Regression	0.785	0.606	0.55	0.674	0.805	151
Decision Tree	0.61	0.418	0.467	0.378	0.569	**206**
Random Forest	0.78	0.551	0.45	0.711	0.794	176
KNN	0.73	0.509	0.467	0.56	0.69	182
LR Tuned (threshold=0.25)	0.705	**0.634**	**0.85**	0.505	**0.805**	**95**
RF Tuned (threshold=0.20)	0.58	0.58	**0.967**	0.414	0.794	**92**

Cost matrix: FN × 5 + FP × 1 | Green = best | Red = worst

Model	Mean Cost	Std
LR	**148.4**	± 26.1
DT	**144.4**	± 21.1
RF	165.8	± 20.5
KNN	166	± 13.9

Model	Mean F1	Std
LR	**0.5042**	± 0.0988
DT	0.4955	± 0.0723
RF	0.4488	± 0.0826
KNN	0.4379	± 0.0540

ROC Curve Analysis



LR **AUC=0.805**

Best - highest discrimination

RF **AUC=0.794**

Strong but lower recall

KNN **AUC=0.690**

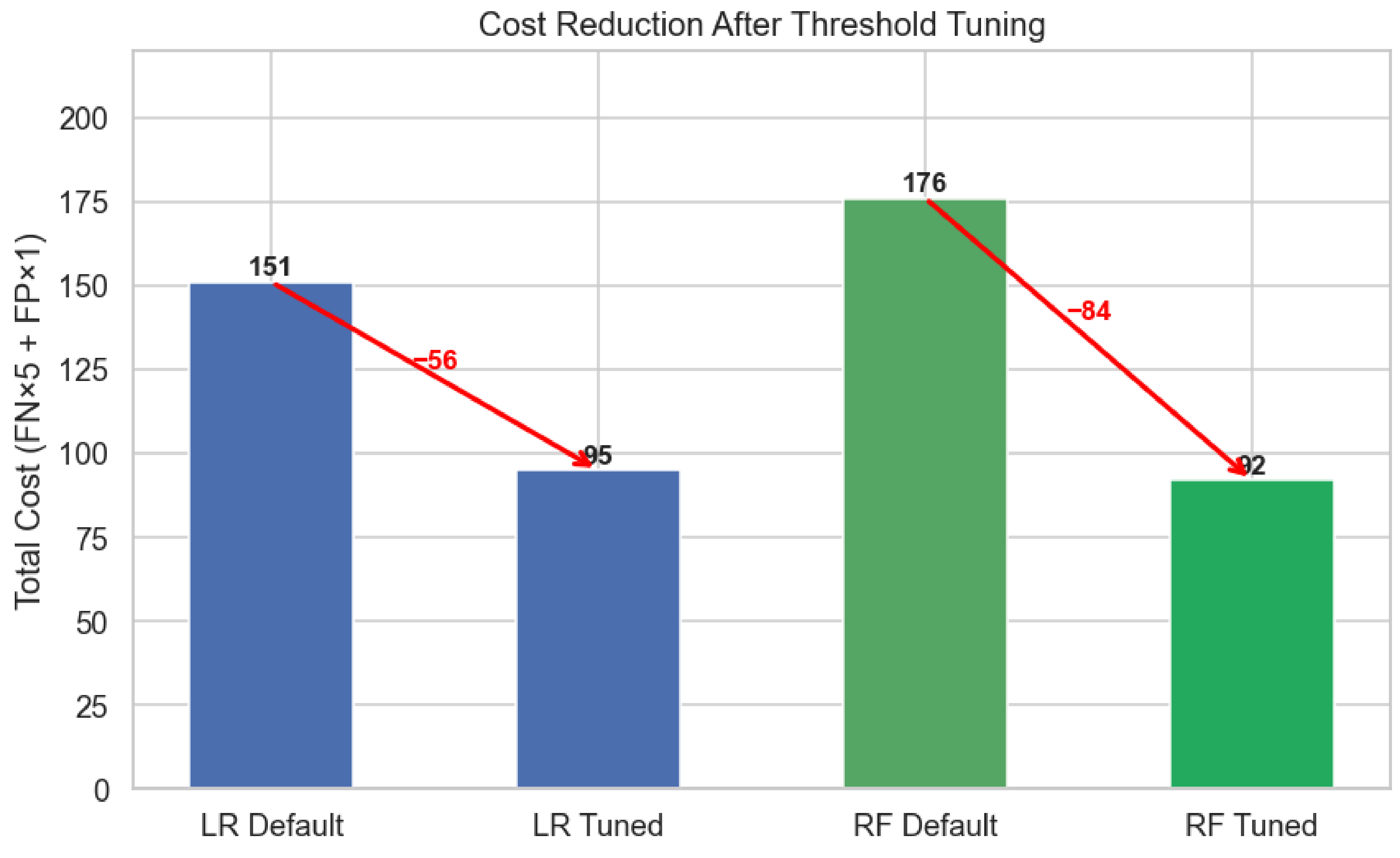
Moderate - not competitive

DT **AUC=0.569**

Weakest -overfitting

Threshold Tuning – Key Finding

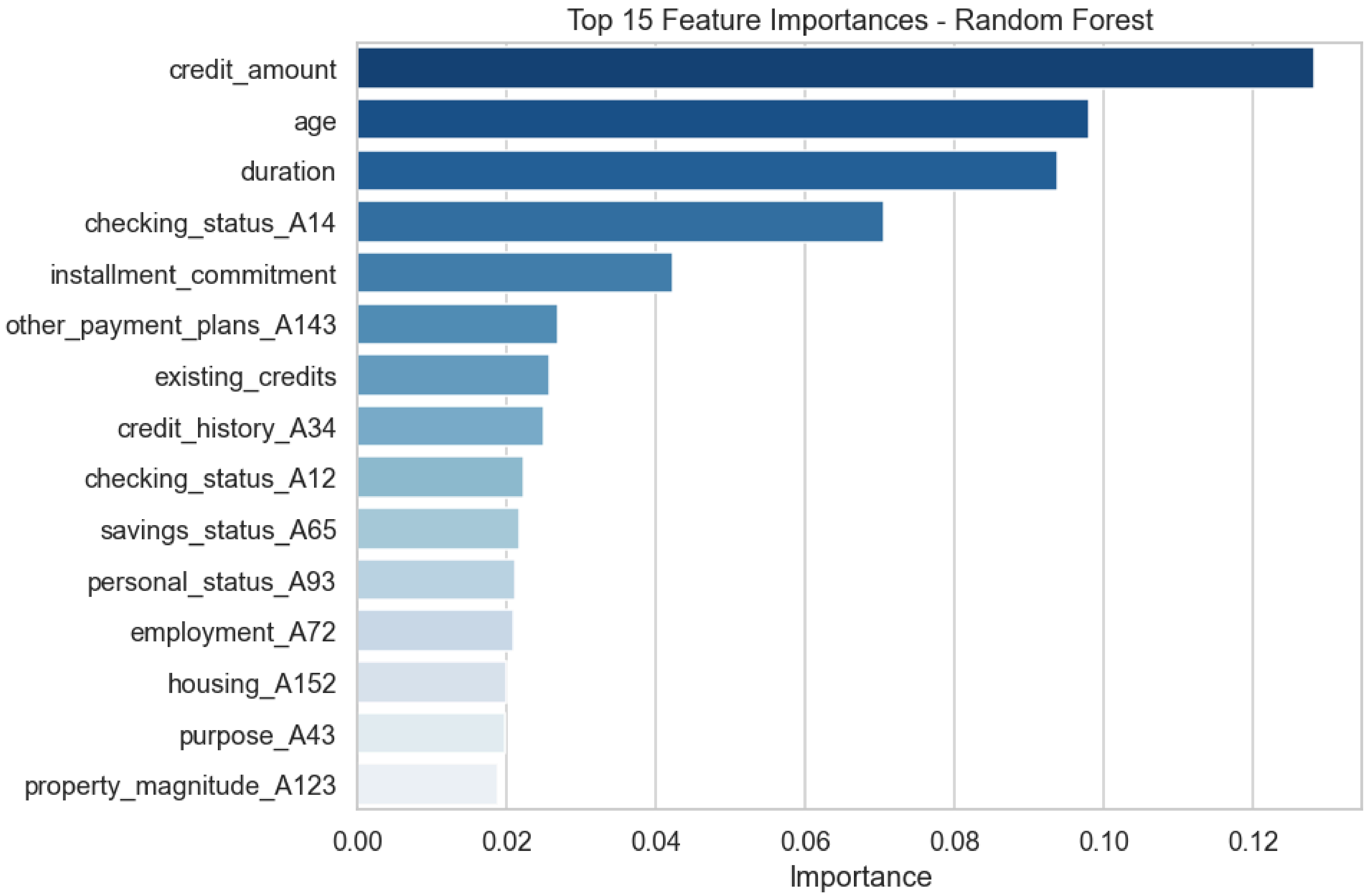
151	**95**	**176**	**92**
LR Default	**LR Tuned**	**RF Default**	**RF Tuned**
threshold = 0.50	threshold = 0.25	threshold = 0.50	threshold = 0.20



Key Insight

- LR: 151 → 95 (37% reduction)
- RF: 176 → 92 (48% reduction)
- LR tuned: 9 bad applicants missed
- RF tuned: only 2 missed but rejects 82 good applicants
- Threshold tuning > model complexity
- Bank risk appetite determines choice

Feature Importance - Random Forest



credit_amount 0.128

Larger loans = higher default risk

age 0.098

Younger = more likely to default

duration 0.094

Longer loans = more exposure

checking_status_A14 0.070

No account = strongest categorical signal

installment_commitment 0.042

Higher rate = more financial strain

Conclusion & Limitations

Best Model

LR Tuned (0.25) - F1 0.634, cost 95. RF Tuned achieves cost 92 but rejects 82 good applicants. Choice depends on bank risk appetite.

Key Finding

Threshold tuning reduces cost 37-48% without changing the model. Evaluation protocol matters more than model complexity.

Feature Insight

credit_amount, age, duration are top 3 predictors. Loan characteristics matter more than personal demographics.

Limitations

- Dataset from 1994 Germany - may not generalise
- IID assumption may be violated
- No fairness analysis on personal_status (gender)
- Single dataset - needs replication

Future Work

- Fairness analysis on demographics
- Replicate on recent credit datasets
- Cost-sensitive learning during training

THANK YOU